Learning Check

1. The temperature of blackbody A is 2,000 K and that of blackbody B is 1,000 K. Regarding the electromagnetic radiation emitted by the two blackbodies:
   (a) A emits more energy per second at any given wavelength than B.
   (b) The peak of A's blackbody radiation curve is at a longer wavelength than that of B's.
   (c) B emits infrared but A does not.
   (d) All of the above.
2. (True or False.) The energy of an oscillating atom in a blackbody can be any value within a certain limited range.
3. When something is restricted to having only certain numerical values, it is said to be ______________.

ANSWERS: 1. (a) **2.** False **3.** quantized

Learning Check

1. Is the energy associated with a photon of blue light
   (a) greater than
   (b) less than
   (c) equal to
   the energy associated with a photon of red light? Why?
2. (True or False.) In the photoelectric effect, increasing the frequency of the light being used will increase the energy of the electrons that are ejected.
3. (True or False.) In the photoelectric effect, increasing the brightness of the light being used will increase the energy of the electrons that are ejected.
4. Name two common devices that make use of the photoelectric effect or similar process in which electrons absorb photons.

ANSWERS: 1. (a), higher frequency **2.** True **3.** False **4.** photocopier, laser printer, digital camera, solar cell, . . .

Learning Check

1. A hot luminous solid emits a ______________ spectrum, whereas a hot luminous gas produces a(n) ______________ spectrum.
2. (True or False.) Two different gases can have the same emission-line spectrum.

ANSWERS: 1. continuous, emission-line **2.** False

Learning Check

1. Which of the following is *not* true in the Bohr model of the atom?
   (a) The electrons orbit the nucleus.
   (b) The orbits of the electrons are quantized.
   (c) While moving in a stable orbit, electrons can emit photons.
   (d) Electrons can jump to a higher-energy orbit if they gain the right amount of energy.
2. In the Bohr model of the atom, a photon may be ______________ when an electron jumps from a large, high-energy orbit to a smaller, low-energy orbit.
3. An atom is said to have been ______________ when it absorbs a photon with sufficient energy to free an electron.
4. (True or False.) There is always a direct correlation between the absorption spectrum of a particular gas and its emission spectrum.

ANSWERS: 1. (c) **2.** emitted **3.** ionized **4.** True

Learning Check

1. If the speed of an electron increases, its de Broglie wavelength
   (a) increases.
   (b) decreases.
   (c) stays the same.
   (d) may increase or decrease.
2. Name a device that makes use of the wavelike nature of electrons.
3. The wavelike aspect of particles is responsible for which of the following?
   (a) The diffraction of electrons as they pass through aluminum
   (b) the quantized orbits of the electron in a hydrogen atom
   (c) The ability of electrons to tunnel through a gap
   (d) All of the above.
4. (True or False.) At any instant in time, it is possible to specify simultaneously the position and the momentum of an electron to arbitrarily high precision.

ANSWERS: 1. (b) **2.** electron microscope, STM **3.** (d) **4.** False

Learning Check

1. Indicate whether the following transitions within the energy-level diagram of hydrogen are accompanied by the *emission* or *absorption* of photons.
   (a) $n = 5$ to $n = 3$
   (b) $n = 1$ to $n = 8$
   (c) $n = 8$ to $n = 9$
   (d) $n = 3$ to $n = 5$
   (e) $n = 6$ to $n = 5$
2. Assuming only one photon is emitted or absorbed in the course of each transition given in Question 1, rank order (from highest to lowest) the frequencies of the photons involved in each transition.
3. The Balmer series is
   (a) a set of play-off matches in golf named after the famous professional golfer Arnold Balmer.
   (b) a sequence of emission lines in the ultraviolet portion of the electromagnetic spectrum from hydrogen.
   (c) a sequence of emission lines in the visible portion of the electromagnetic spectrum from hydrogen.
   (d) a sequence of emission lines from hydrogen involving transitions from higher levels down to the $n = 3$ energy level.
4. What exactly is (or are) "excluded" by the Pauli exclusion principle?
5. (True or False.) The chemical properties of the elements in the periodic table are primarily determined by the electronic ground-state configuration of their atoms.

ANSWERS: 1. Emission (a), (e) Absorption (b), (c), (d). **2.** (b), (a) & (d), (e), (c) **3.** (c) **4.** Electrons in the same quantum state. **5.** True.

Learning Check

1. (True or False.) An x-ray can be emitted only when an electron undergoes an allowed energy-level transition.
2. Rank (from lowest to highest) the frequencies of the characteristic K_α radiation emitted by the following elements:
   (a) iron ($Z = 26$)
   (b) zinc ($Z = 30$)
   (c) titanium ($Z = 22$)
   (d) copper ($Z = 29$)
3. Which of the following processes or phenomena is *not* involved in the operation of lasers?
   (a) stimulated emission
   (b) population inversion
   (c) metastable state
   (d) blackbody radiation
4. Which of the following common devices does *not* employ a laser in its operation?
   (a) DVD player
   (b) supermarket checkout price scanner
   (c) telephonic fiber-optics communication system
   (d) plasma TV
5. ______________ are three-dimensional images made with the aid of lasers.

ANSWERS: 1. False 2. (c), (a), (d), (b) 3. (d) 4. (d) 5. Holograms